If your goal is to **reach PhD-level AI researcher capability without actually doing a PhD**, the path is very real—but it's **not casual learning**. You'll essentially replicate what top universities like MIT, Stanford University, or Carnegie Mellon University do—just in a **self-directed, structured, and output-driven way**.

Given your background (18 yrs engineering + architecture), you actually have a **massive advantage**—you can go deeper into **systems + applied research + infra**, which many PhDs lack.

🚀 Informal "PhD-equivalent" AI Research Path

🧠 Phase 0: Mindset Shift (Most Important)

You are NOT:

- "learning AI"

You ARE:

- **producing new knowledge**
- **reading papers daily**
- **building novel systems**
- **thinking in math + abstractions**

👉 Think like:

> "Can I improve this paper?" not "Can I understand this?"

📚 Phase 1: Math Foundations (3–6 months, deep not wide)

Core Areas

- Linear Algebra (vectors, eigenvalues, SVD)
- Probability & Statistics
- Optimization (gradient descent, convexity)
- Information Theory

Key Resources

- MIT OpenCourseWare (Linear Algebra – Gilbert Strang)
- Pattern Recognition and Machine Learning
- Deep Learning

👉 Output requirement:

- Derive backprop from scratch
- Implement optimization algorithms from scratch (SGD, Adam)

🤖 Phase 2: Core ML + DL Systems (3–6 months)

Learn like a researcher:

Not “use PyTorch” But:

- “Why does this converge?”
- “What breaks?”

Topics

- Classical ML (SVM, trees, Bayesian models)
- Neural networks (MLP, CNN, RNN)
- Transformers (attention, scaling laws)

Code-level mastery

- Rebuild:
  - GPT (decoder-only transformer)
  - Training loop
  - Loss functions

Key Tools

- PyTorch

- Hugging Face Transformers

👉 Output:

- Write your own mini GPT (no libraries except tensor ops)
- Train on small dataset

🔬 Phase 3: Paper Reading & Reverse Engineering (ONGOING)

This is where you become "PhD-level".

Start With:

- arXiv
- Papers from:
  - OpenAI
  - DeepMind
  - Google Research

Process (CRITICAL)

For each paper:

1. Read abstract + diagrams
2. Re-derive math
3. Re-implement
4. Improve something

👉 Output:

- "paper reproduction repo"
- blog explaining paper like researcher

🧪 Phase 4: Choose Research Track (VERY IMPORTANT)

You must specialize like a PhD.

Options:

1. LLM / Generative AI (your current direction)

- Transformers
- RLHF
- Alignment
- Scaling laws

2. Systems for AI (VERY STRONG FOR YOU)

- Distributed training
- GPU optimization
- inference scaling
- model serving

3. Applied AI

- Healthcare / Finance / Industrial AI

👉 Suggestion for YOU: **LLM + Systems (rare + high impact combo)**

🧱 Phase 5: Build Research Portfolio

This replaces your "PhD thesis".

Must-have artifacts:

- 3–5 serious research projects
- 1–2 arXiv-style papers
- GitHub with deep repos
- Technical blog (very important)

👉 Example projects:

- Improve attention efficiency
- Build distributed LLM training system
- Novel tokenizer or embedding method

📄 Phase 6: Write & Publish (This is what makes you “PhD-level”)

You don’t need university if:

- You **publish**
- You **contribute to research**

Platforms:

- arXiv
- Papers With Code
- Conferences (long-term):
  - NeurIPS
  - ICML
  - ICLR

🧑‍🔬 Phase 7: Join Research Ecosystem

You need peers (very important)

- Twitter/X AI community
- Discord research groups
- Open-source contributions

Target roles:

- Research Engineer
- Applied Scientist
- Independent Researcher

⚡ Realistic Timeline (Aggressive)

Phase	Duration
Math + ML foundations	6 months
Deep systems + transformers	6 months
Paper-level research	6–12 months
Publish & recognition	1–2 years

👉 Total: **1.5–3 years (if serious)**

🔥 What Differentiates You from Typical Learners

You already have:

- Architecture thinking ✅
- System design ✅
- Engineering maturity ✅

👉 Add:

- Math depth
- Research mindset
- Paper writing

= You become **Research Engineer / AI Architect (rare profile)**

🧭 Practical Weekly Routine (PhD-style)

- 2 days → math + theory
- 2 days → coding + experiments
- 2 days → paper reading + reproduction
- 1 day → writing (blog / notes)

⚠️ Reality Check

This path works ONLY if:

- You **build + publish**, not just watch courses
- You **struggle with papers daily**
- You **create something new**

Otherwise, you'll stay at:

> "Engineer using AI" (not "AI researcher")

💡 Final Insight

A PhD is essentially:

> Structured suffering + guided research + publishing

You can replicate all 3:

- Structure → roadmap (above)
- Guidance → papers + open-source
- Output → GitHub + arXiv

If you want, I can:

- Design a **12-month exact roadmap (week-by-week) tailored to YOU**
- Give **project ideas that can actually get published**
- Or help you **pick a research niche where you can stand out fast**